Introduction to linux

Plan

- **What is _ Linux?**
- **What is _ GNU/Linux?**
- **What is a distribution?**

What is _ Linux?

- Linux is a kernel created by Linus Torvald in 1991
  - https://github.com/torvalds
- The Linux kernel is an operating system kernel UNIX.
- It is developed by hundreds or thousands of volunteers around the world : the community
  - The development process is public on Internet
  - The development cycle is fast and incremental
- The release of the stable **5.19.12** LTS version of the Linux kernel was announced on 2022-09-28 by Linus Torvalds.

What is _ GNU/Linux?

- The GNU Project started in 1984 with richard stallman
- The project is supported by the Free Software Foundation
- The idea is to provide a complete and completely free operating system.
- The project is maintained by a community of hackers
- The GNU Project provides a large suite of free tools, but no kernel
- GNU / Linux is a totally Free and powerful operating system and a large suite of GNU tools on one kernel, the Linux kernel. GNU on Linux hence the slash

What is a distribution

- A distribution is mostly :
  - A set bringing together the components of a system in a coherent and stable whole whose installation, use and maintenance are facilitated. They therefore most often include installation software and configuration tools.

The basics of shell

Introduction to the shell

bash

Plan

- The Shell- Definition
- Why the Shell ?
- The different Shell
- Access to the Shell

The Shell- _ Definition

- The Shell is a program that provides an interface in text mode
  1. A command line interface – CLI

  2. A language of programming

Why the Shell ?

- More Powerful than a GUI: Many actions impossible to perform in GUI
- More practical: Allows automation, many keyboard shortcuts, completion…
- More accessible – few resources (memory, cpu), access (no MISTLETOE)
- Simple programming of actions
- Has many internal commands (cd, ls, ..) no process
- documentation difficult to access for the beginner
- The syntax is not easy for the beginner

The different Shell

- Stephen 's Shell Bourne
  - Bourne shell (sh)
    - History
  - Bourne-Again shell (bash)
    - The Shell GNU
- David 's Shell Korn
  - Kornshell (ksh)
- Csh: C shell
  - tcsh (modern version of csh)
    - Syntax very close to the language VS
- Zch : Z Shell
  - Taking the most practical functions of bash, ksh and tcsh
    - Correction spelling

Access to Shell

- In terminal mode , no GUI started :
  - Log in , the Shell is executed
  - Virtual consoles : ALT + fx
- In GUI mode :
  - Launch an emulator terminal
    - Gnome-terminal, xterm, …

Basics of shell

Understand the syntax of a command

Plan

- The prompt of commanded
- The syntax of commands

The prompt

- Prompt
  - When connecting or launching a terminal emulator, the prompt appears :
    - The prompt $ for a user and # for root.

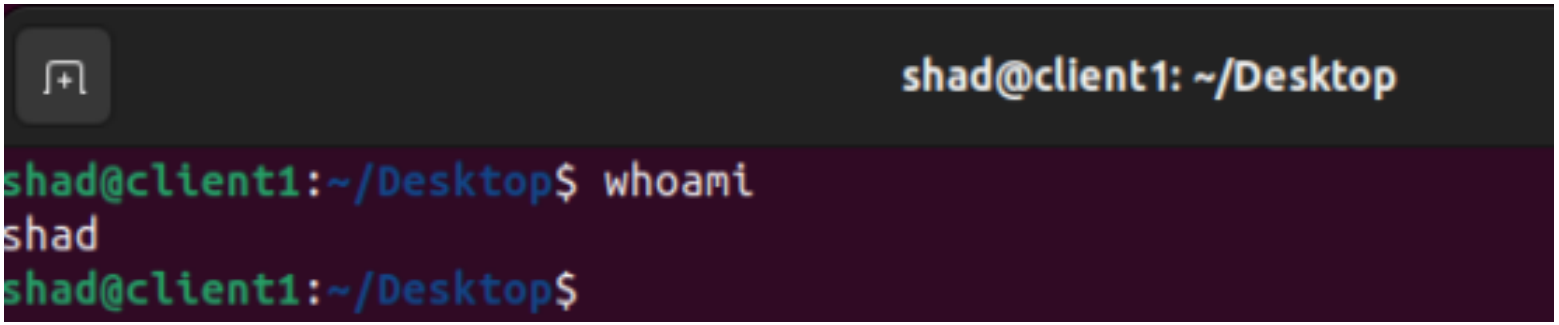
```
username@machine ~ $

root@machine ~ #
```

The syntax of a command

name _cmd	*: the name of the command*
option	*: command options*
argument	*: command parameters (file , directory)*
Space	*: Field separator, separates the elements*

whoami

- The **whoami** command is used to display the username you have used to log in to the system. Sometimes, you may need to log in to a system and switch among different users. If you lose track of which account you are logged into, use the **whoami** command.



hostname

- The **hostname** command is used to display the hostname of the system you are currently logged in to. When you log in to different systems using the same terminal, you can use the hostname command to find the system on which you are presently running the command.

```
shad@client1: ~/Desktop
shad@client1:~/Desktop$ hostname
client1
shad@client1:~/Desktop$
```

date

```
shad@client1: ~/Desktop
shad@client1:~/Desktop$ date
Tue Apr 11 11:20:47 AM EDT 2023
shad@client1:~/Desktop$
```

date +[format]

%D	Displays the date in the format mm/dd/yy, where mm is month, dd is day, and yy is year.
%H	Displays the current hour in the 24-hour format (00 to 23).
%S	Displays the seconds (00 to 60).
%T	Displays the time in the 24-hour format; i.e., hh:mm:ss. This choice displays the seconds also.
%y	Displays the last two digits of the current year.

```
shad@client1: ~/Desktop
shad@client1:~/Desktop$ date +%Y
2023
shad@client1:~/Desktop$
```

uptime

- The **uptime** command displays the time from when a system started running. The output of the uptime command gives information about the current time, how long the system is running, and how many users are currently logged in.
- The last field of the **uptime** command output displays the system's load averages for the last 1 minute, 5 minutes, and 15 minutes. This information can be used to check whether the system is busy.

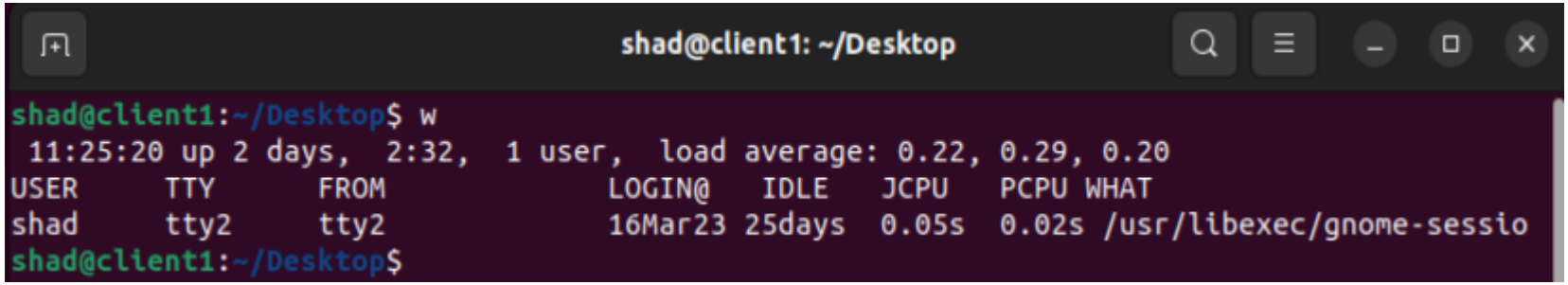
who

- If you're working on a server, there may be multiple users logged in at the same time. The **who** command is used to figure out the details of users currently logged in to a system. The output of the who command includes the username, the name of the system from which the user is connected, and the time since the user is connected.

```
shad@client1: ~/Desktop
shad@client1:~/Desktop$ who
shad     tty2         2023-03-16 12:54 (tty2)
shad@client1:~/Desktop$
```

W

- The **w** command is primarily used to display the details of users who are currently logged in to a system and their transactions. The first line of the output displays the status of the system. The second line of the output displays a table with the first column listing the users logged in to the system and the last column showing the current activities of the users. The remaining columns of the table show different attributes associated with the users.



The history of commands

- The .bash_history file keeps the last 1000 commands
  - The history command (history: displays the last 10 commands)
  - The arrows : UP and DOWN to walk around history
  - Search in history with CTRL+R (lowercase) + a string of temper

The completion

- With the key TAB, the Shell completes
  - the name of files
  - the name of commands

Keyboard Shortcuts

ctrl+a	place the cursor at the beginning of line
C t r l +e	place the cursor at the end of the line
C t r l +w	Deletes the word behind the cursor
Ctrl+k	Deletes all the words in front of the cursor
Ctrl+x and Backspace	Deletes all the words behind the cursor
C t r l +c	Kills the command by Classes
ctrl+d	Exit the current Shell (equivalent to "exit ”)
ctrl+l	Clean your screen (equivalent to “clear ")

what we have covered

- The history of commands
- The completion
- Keyboard Shortcuts